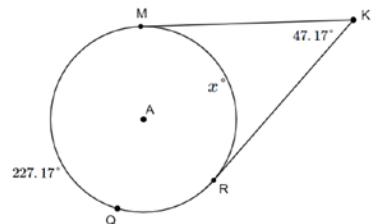


For questions 1-2, Circle A has points M , Q , and R on the circle. Tangents $\overline{MK}$ and $\overline{RK}$ intersect at point K outside the circle. The $m\widehat{MQR} = 227.17^\circ$, $m\widehat{MR} = x^\circ$, and $m\angle MKR = 47.17^\circ$.



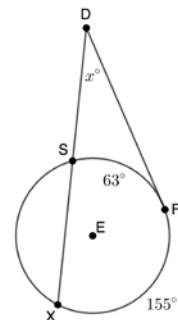
1. Write a formula that can be used to find the value of x .

$$x = 360 - 227.17 \text{ or } \frac{227.17 - x}{2} = 47.17$$

2. Find the value of x .

$$x = 132.83$$

For questions 3-4, Circle E has points X , S , and F on the circle. Tangent $\overline{FD}$ and secant $\overline{XD}$ intersect at point D outside the circle. The $m\widehat{XF} = 155^\circ$, $m\widehat{SF} = 63^\circ$, and $m\angle SDF = x^\circ$.



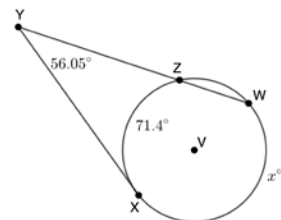
3. Write a formula that can be used to find the value of x .

$$x = \frac{155 - 63}{2}$$

4. Find the value of x .

$$x = 46$$

For questions 5-6, Circle V has points X , Z , and W on the circle. Tangent $\overline{XY}$ and secant $\overline{WY}$ intersect at point Y outside the circle. The $m\widehat{XW} = x^\circ$, $m\widehat{XZ} = 71.4^\circ$, and $m\angle XYZ = 56.05^\circ$.



5. Write a formula that can be used to find the value of x .

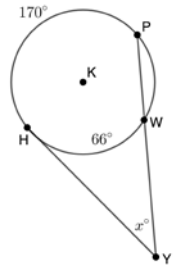
$$\frac{x - 71.4}{2} = 56.05$$

6. Find the value of x .

$$x = 183.5$$

7. Circle K has points P , W , and H on the circle. Tangent $\overline{HY}$ and secant $\overline{PY}$ intersect at point Y outside the circle. The $m\widehat{HP} = 170^\circ$, $m\widehat{HW} = 66^\circ$, and $m\angle HYW = x^\circ$. What is the $m\angle HYW$?

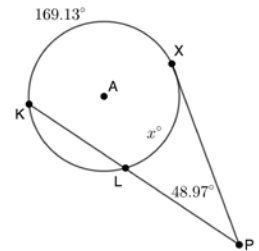
$$x = \frac{170 - 66}{2} = m\angle HYW = 52^\circ$$



8. Circle A has points X , L , and K on the circle. Tangent $\overline{XP}$ and secant $\overline{KP}$ intersect at point P outside the circle. The $m\widehat{KX} = 169.13^\circ$, $m\widehat{XL} = x^\circ$, and $m\angle XPL = 48.97^\circ$. What is the $m\widehat{XL}$?

$$\frac{169.13 - x}{2} = 48.97$$

$$x = m\widehat{XL} = 71.19^\circ$$



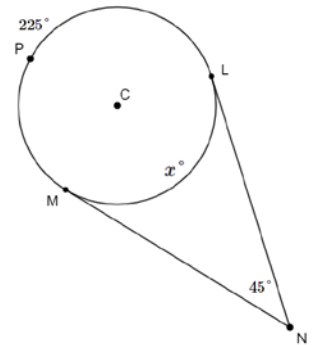
9. Circle C has points L , P , and M on the circle. Tangents $\overline{LN}$ and $\overline{MN}$ intersect at point N outside the circle. The $m\widehat{LPM} = 225^\circ$, $m\widehat{LM} = x^\circ$, and $m\angle LNM = 45^\circ$. What is the $m\widehat{LM}$?

$$x = 360 - 225 = m\widehat{LM} = 135^\circ$$

or

$$\frac{225 - x}{2} = 45$$

$$x = m\widehat{LM} = 135^\circ$$



10. Circle O has points G , J , and K on the circle. Tangents $\overline{GH}$ and $\overline{JH}$ intersect at point H outside the circle. The $m\widehat{GK} = 249^\circ$, $m\widehat{GJ} = 111^\circ$, and $m\angle GHJ = x^\circ$. What is the $m\angle GHJ$?

$$x = \frac{249 - 111}{2} = m\angle GHJ = 69^\circ$$

